

Use Case “Place Order”

1. Use case code

UC00X

2. Brief Description

This use case describes the interaction between Customer and Aims system when Customer places an order for products in the cart and proceeds to payment.

3. Actors

a. Customer

4. Preconditions

a. The Aims system is running normally

b. Products in the cart exist in the system

c. There is at least one item in the cart.

5. Basic Flow of Events

1. Customer requests to place an order from the cart
2. The system checks whether the available stock is sufficient to fulfill the customer's demands
3. If the stock is sufficient, the system displays the delivery information form together with the cart information
4. Customer enters and submits delivery information
5. The system validates the delivery information and asks the customer to update if in case any required field is blank or invalid
6. Each time the customer enters or updates the delivery address, the system calculates or recalculates the shipping fee
7. The system displays and temporarily saves the invoice information, including the list of products in the cart, quantity, product prices, total product price excluding VAT, total product price including VAT, delivery fee, and total amount need to be paid
8. Customer requests to pay for the order
9. The system continues with the use case Pay Order
10. After successful payment, the system displays the general order information, including customer name, phone number, shipping address, province, and total amount, together with transaction information including transaction ID, transaction content, and transaction time
11. The system clears the cart
12. The system sets the order status to Pending Processing
13. The system sends the invoice and payment transaction information to the customer's email
14. The system records the payment transaction and successful order information

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	One or more products in the cart do not have sufficient stock to fulfill the customer's requested quantity	The AIMS software notifies the customer that some products in the cart are out of stock or insufficient. Customer is required to update the cart	Use case ends and returns to View Cart
2.	At Step 5	The delivery information entered by the customer is incomplete	The system displays an error message indicating the invalid or missing information. The customer is requested to re-enter valid delivery information	Step 3: Display delivery information form)
3.	At any step	The customer want to go back to a previous screen or exits the ordering process	- The system saves the current cart information - The ordering process is canceled or paused The ordering process is canceled or paused	Use case ends
4.	At step 9	Payment fails or the customer cancels the payment while it is in process	- The system displays a message that Payment is unsuccessful - The customer may retry the payment or modify the order	Step 7: displays invoice information

7. Input data

Table A-Input data of place order

No	Data fields	Description	Mandatory	Valid condition	Example
1.	String	Name of the person who will receive the order	Yes	Not empty	Bùi Doãn Khang
2.	Numeric	Phone number of the receiver used for delivery contact	Yes	- Not empty - Contain 10 digits and start with '0'	0123456789 0123 456 789

3.	String	Province or city where the order will be delivered	Yes	Must be selected from the list of supported provinces/cities in Vietnam	Hà Nội Hồ Chí Minh city
4.	String	Detailed address where the order will be delivered	Yes	Not Empty	1 trần đại nghĩa street, hà nội
5.	String	Some notes for the delivery process.	No	Not exceed 255 characters	Please call before delivery or send message

8. Output data

Table B-Output data of Place Order

No	Data fields	Description	Display format	Example
1.	String	Title of the media product displayed in the invoice	Text	Books: master machine learning
2.	Integer	Price of a product in the order (excluding VAT)	Positive number with thousands separator	120,000
3.	Integer	Quantity of the product in the order	Positive integer	2
4.	Integer	Total price for each product (Product Price × Quantity)	Positive number with thousands separator	1,000,000
5.	Integer	Total price of all products in the order before VAT is applied	Positive number with thousands separator	5,000,000
6.	Integer	Total product price including VAT	Positive number with thousands separator	5,200,000
7.	Integer	Shipping fee calculated by the system based on delivery address and product weight	Positive number with thousands separator	40,000
8.	Integer	Total payment amount including product price (include VAT) and delivery fee	Positive number with thousands separator	5,240,000

9.	String	Name of the person receiving the order	Text	Bùi Doãn Khang
10.	Integer	Phone number of the order receiver	Numeric Text	0123456789
11.	String	Delivery address of the order	Text	1 trần đại nghĩa street, hà nội
12.	String	Province or city of the delivery address	Text	Hà Nội
13.	String	ID of the payment Transaction	Numeric Text or/and Text	ABC123
14.	String	Description or content of the payment transaction	Text	Payment for ABC123
15.	Datetime`	Date and time when the payment transaction occurred	dd/MM/yyyy HH:mm:ss	1/1/2026 12:30:00

9. Postconditions

Success Postcondition:

- The place order state is successful
- A new order is created in the system
- The order status is set to Pending Processing
- The system records the order information and payment transaction information
- The customer's cart is cleared
- The system sends the invoice and payment transaction information to the customer's email

Failure Postcondition

- No order is created in the system
- No payment transaction is recorded
- The products remain in the customer's cart so that the customer can modify the order or attempt payment again